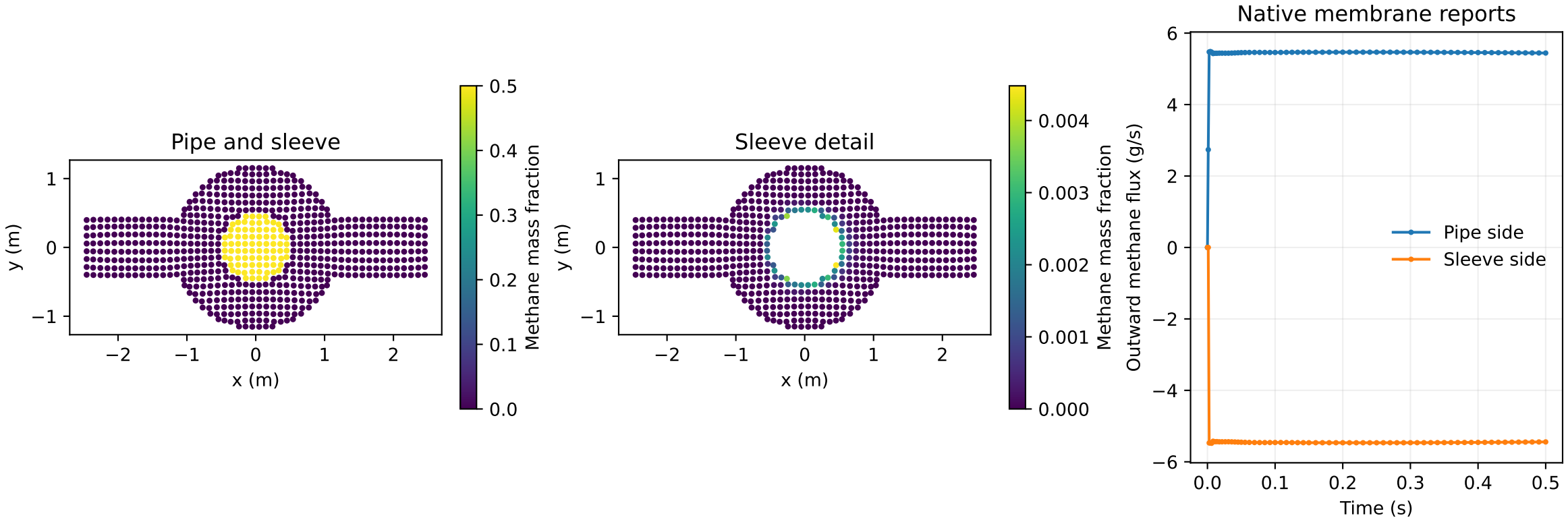**Selective methane transport through a membrane**